Technical Service Bulletin

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Premium K STC Injectors (Field Fix 177)	

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Warranty Statement

The information in this document has no effect on present warranty coverage or repair practices, nor does it authorize TRP or Campaign actions.

Contents

This document was originally released between 1994 and 2001. It has been added to QSOL for informational purposes

This document announces the release of a new Premium K STC injector. The new Premium K STC injector assemblies will supersede and obsolete all Standard K and Family K STC injector assemblies. The features of the Premium K STC injector are: a barrel and plunger assembly with press fit plunger-to-coupling joing, a longer barrel, a stronger cup, a new oil feed lock nut, a new top-stop screw, and a ceramic internal link (in most injectors). The Family K lock nut, a new top-stop screw, and a ceramic internal link (in most injectors). The Family K injector do have the same longer barrel and stronger cup featured in the Premium K injector, but do **not** have the other features. Premium K injector Part Nos. 3076700, 3076702, and 3076703 are **not** manufactured with a ceramic internal link. If the internal link needs to be replaced these three injectors during rebuild, use a ceramic link.

The Premium K STC injector is more reliable and durable than the Standard K and Family K STC injector, unless the latter two injector types have been upfit with a ceramic internal link and a barrel and plunger assembly with a press fit coupling-to-plunger join.

Refer to Table 1 for new Premium k injector assembly part numbers.

Table 1 - Injector Part Numbers, Engine Models, Control Parts List

New Injector Part No.	Old Injector Part No.	Engine Model	CPLs
3076130	3062092 3062093	KTTA/KTA-19	951, 1253, 1316, 1319, 1430, 1499
3076132	3058802 3058803 3053206* 3053207*	KTTA-19 KTTA/KTA-38 KTTA/KTA-50	816, 841, 844, 846, 859, 860, 872, 873, 876, 878

New Injector Part No.	Old Injector Part No.	Engine Model	CPLs
3076133	3056495	KTTA-50	968
3076134	3279777 3058800 3058801 3052226* 3052227*	KTTA/KTA-19	569, 720, 739, 756
3076700	3059927	KTTA-19	1170
3076702	3067393	KTTA/KTA-50	1254, 1528, 1544
3076703	3279719	KTA-38 KTA-50	1219, 1251, 1495, 1496, 1497, 1498, 1541, 1542, 1543
3076704		KTTA-50	1250
3077714	3059458 3059459	KTA-19	0801
3077715	3279847	KTA-19	1455, 1457
3077760		KTA-38	1628
3627922		KTA-50	1602
*These are Hydraulic Variable Timing (HVT) injectors. When replacing them with Premium K STC injectors, the existing oil transfer connection **must** be replaced with a different connection. See the injector section in the appropriate parts catalog for the correct part number.			

The press-fit plunger assemblies have a stronger joint between the coupling and the plunger than the older crimped assemblies. This will improve the plunger to coupling joint.

The ceramic link will provide the same improved internal wear characteristics that were realized in the NT STC injector assemblies with ceramic links.

The barrel is longer to provide improved barrel and plunger life.

Conversion to Premium K STC

A standard K STC injector can **not** be converted to a Premium K STC injector. Very few of the parts are interchangeable. The full injector assembly **must** be replaced.

To convert a Family K STC injector to a Premium K STC injector, replace the oil feed lock nut, the top-stop screw, the internal link (to ceramic), and the barrel and plunger assembly (to one with a press fit coupling-to-plunger join). If the Family K STC injector already has a ceramic internal link and a press fit coupling-to-plunger joing, only the oil feed lock nut and the top-stop screw need to be changed.

Note : No reliability or durability improvement will be gained if the Family K STC injector has previously been upfit with a ceramic internal link and a barrel/plunger assembly with a press fit coupling-to-plunger joint.

Refer to Table 2 for the latest Premium K injector part numbers. Part numbers with a dash indicate it is specific to a particular injector and **not** common to all Premium K injectors. Refer to the Injector Specifications Manual Bulletin No. 3379664 for the correct part number.

Parts Information

Refer to Table 2 for the parts list and Figure 1 for the exploded view of the Premium K injector.

Table 2 - Premium K STC Injector Components

Ref No. (Fig. 1)	Component Part No.	Qty.	Component Description	Vs. Present Standard K STC Injector
1	3068859	1	STC Tappet Top Cap	Same
2	3279850	1	STC Oil Feed Lock Nut	Different
3	3052233	1	STC Tappet Top Link	Same
4	3052231	1	Top Link Retaining Clip	Same
5	3075381	1	STC Tappet	Same
6	3279819	1	Internal Plunger Link	Different
7	3068860	1	Top-Stop Lock Nut	Same
8	3003682	1	Hold Down Clamp	Same
9	3279821	1	Injector Top-Stop Screw	Different
10	---	1	Barrel and Plunger Assy.	Different
11	3052218	1	Spring Retainer	Same

Ref No. (Fig. 1)	Component Part No.	Qty.	Component Description	Vs. Present Standard K STC Injector
12	3042428	1	Compression Spring	Different
13	3042427	1	Compression Spring	Different
14	3042425	1	Injector Adapter	Different
15	174299	1	Filter Screen Retainer	Same
16	3008706	1	Filter Screen	Same
17	-----	1	Orifice Plug (Drilled)	Same
18	173085	1	Orifice Plug (Non-Drilled)	Different
19	173006	2	Orifice Plug Gasket	Same
20	167157	1	Barrel Check Ball	Same
21	203426	2	Roll Pin	Same
22	193736	3	O-Ring Seal	Different
23	-----	1	Cup	Different
24	3042430	1	Cup Retainer	Different

Premium K STC Injector - Parts Explosion

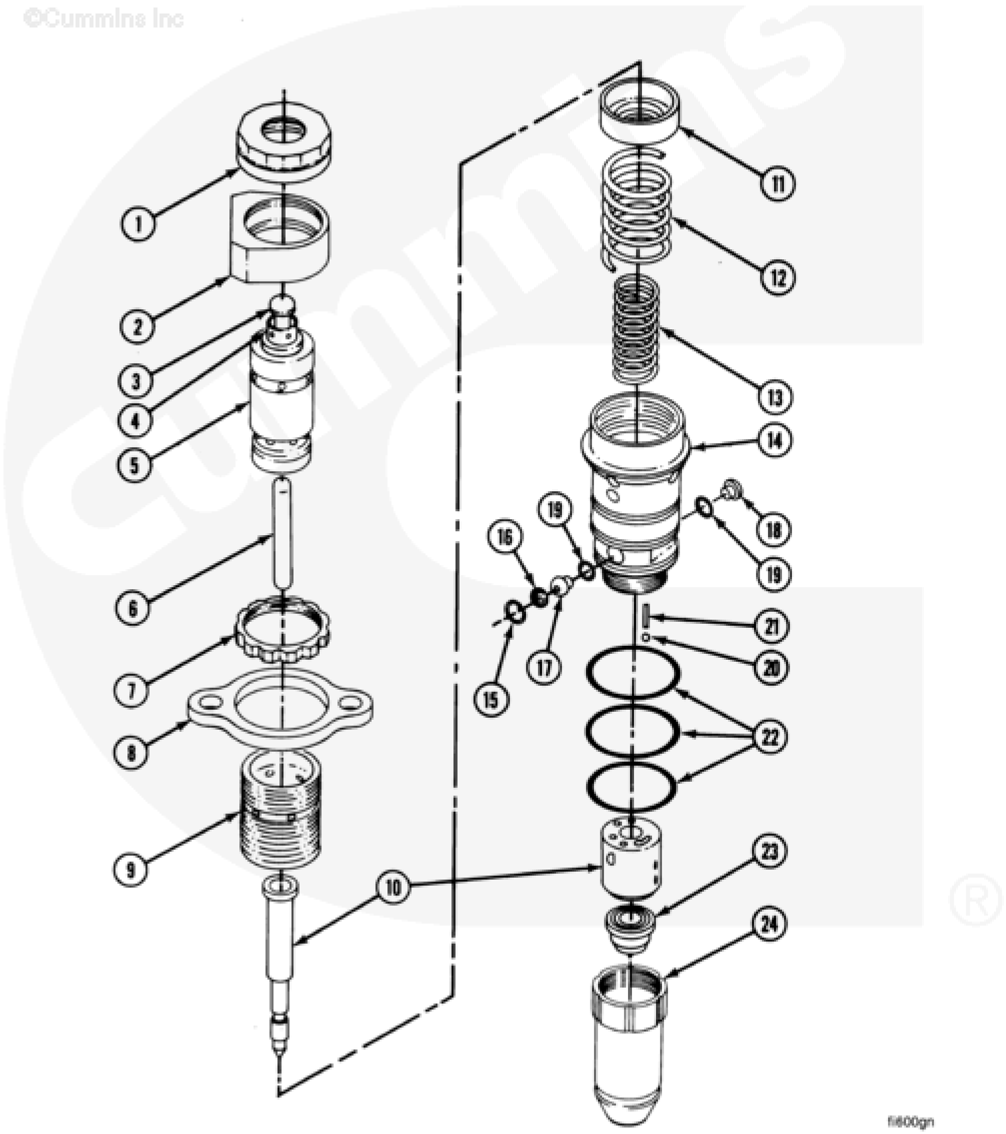

Figure 1

Assembly and Calibration

Assemble the injector using the correct components listed in Tables 2 and 3. Refer to the STC Injector Shop Manual, Bulletin No. 3810313, for assembly and calibration instructions.

Note : The flow calibration of Premium K STC injectors, with top stop travel of 0.393 inches, must be performed before setting the top stop plunger travel.

The Premium K injectors can only be flowed on Hartridge injector stands, Part No. 3375317. They can **not** be flow calibrated on Part No. ST-790.

Use adapter pot, Part No. 3376862 and the 11 ring link, Part No. 3376308 when flow calibrating Premium K STC injector assemblies.

⚠ CAUTION ⚠

Use of the incorrect o-ring on the injector body can cause severe damage to the engine.

The Premium K STC injector used o-ring, Part No. 193736 in all three body o-ring locations.

The Premium K STC injector will have a new oil feed lock nut and top stop screw. The oil feed lock nut is marked with 0.493 or 0.404 inches to indicate the respective total travel settings for the injector. The correct total travel **must** be showing on the oil feed lock nut while the injector is installed in the engine. Refer to Figures 2 and 3.

Note : The oil feed lock nut must be installed so that the correct total travel setting number is facing upward, or excess wear will occur on the lock nut and transfer tube. This wear can cause excess oil leakage which causes incorrect injector timing, misfire, and white smoke.

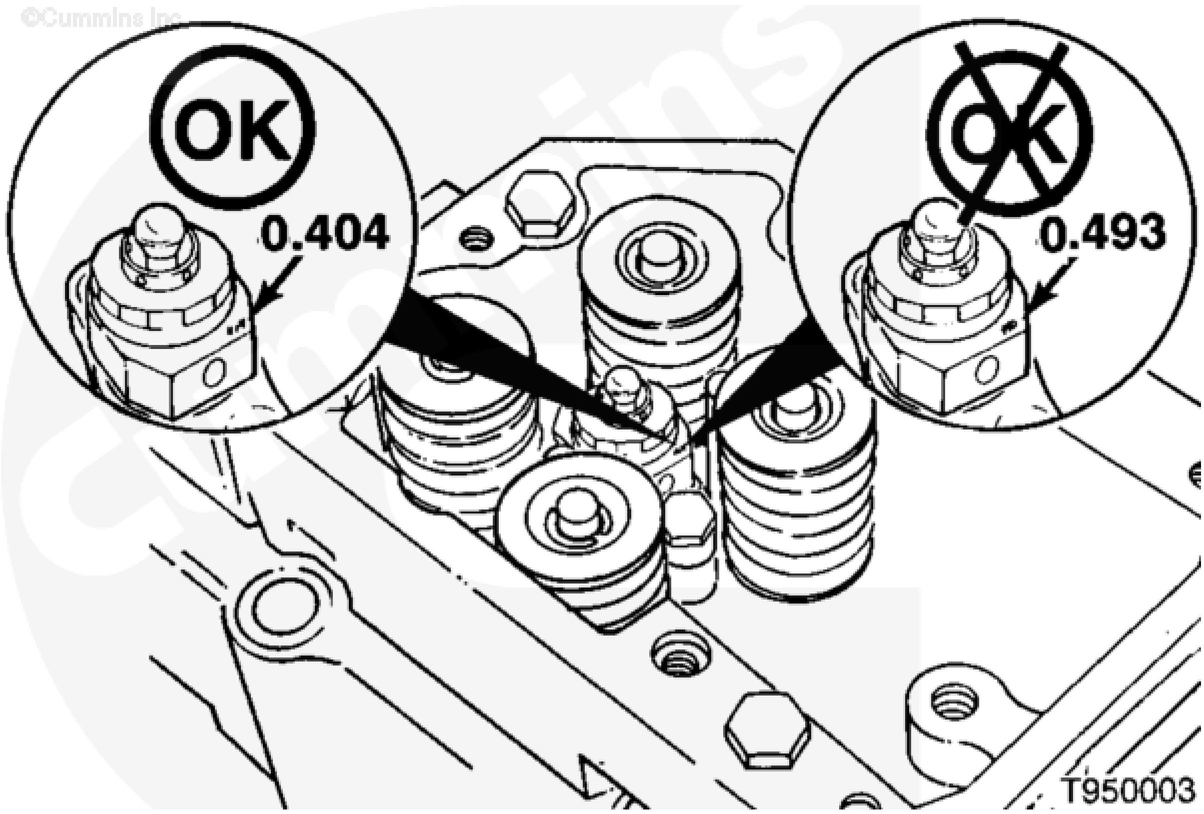

Figure 2 Injector Total Travel = 0.404 inches

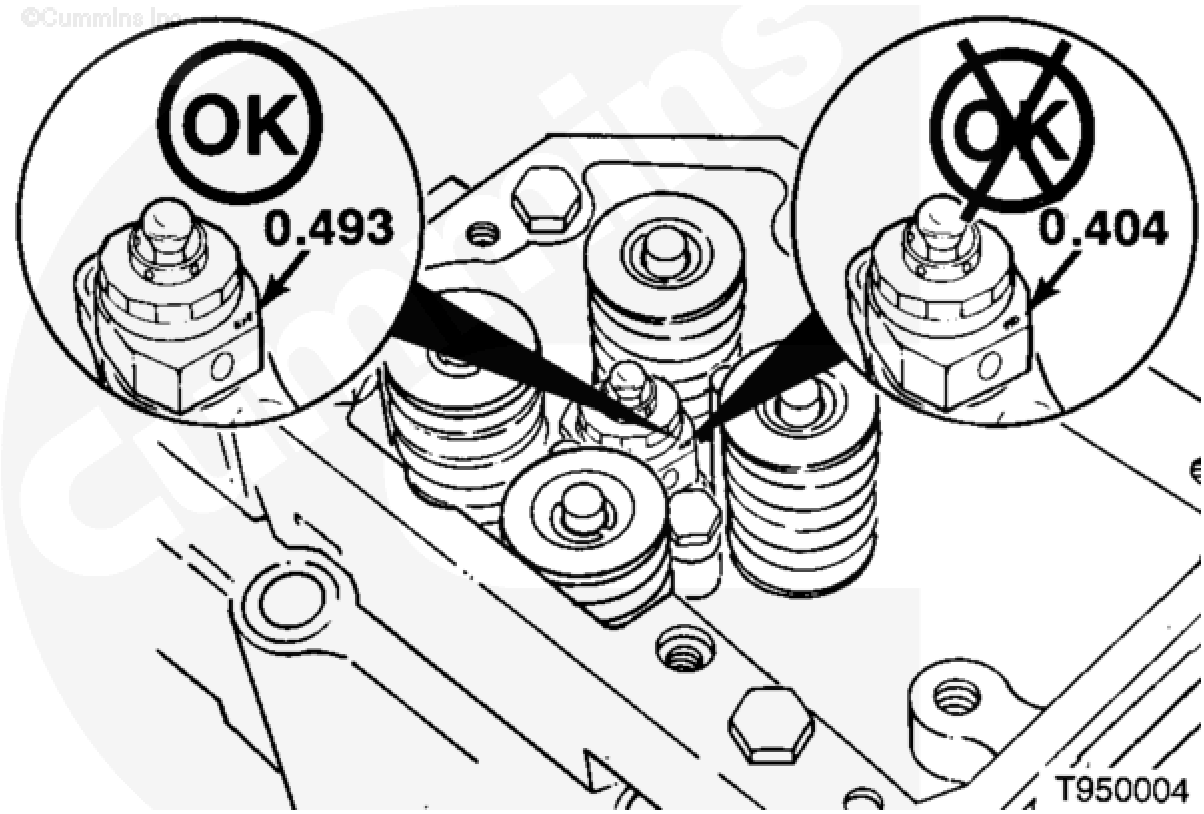

Figure 3 Injector Total Travel = 0.493 inches

Table 3 - Premium K Injector Parts and Flow

Injector Less Link	Barrel Plunger Assy.	Refer. Barrel Part No.	Top Stop Set.	Total Travel	Cup Part No.	Cup Holes No. - Size x Angle	Approx. Injector Orifice	3375317 Delivery mm^3 /Stroke	Cam
3076130	3076125	3053420	0.302	0.404	3279720	10-.0095x 10deg	N/A	340	310
3076132	3076125	3053420	0.302	0.404	3042713	10-.0085x 10deg	0.038	365	310
3076133	3076125	3053420	0.302	0.404	3042712	10-.0101x 10deg	N/A	410	310
3076134	3076127	3063012	0.302	0.404	3279810	10-.0085x 10deg	0.038	365	310
3076700	3076126	3053422	0.393	0.493	3059929	10-.0104x 10deg	N/A	275	426

Injector Less Link	Barrel Plunger Assy.	Refer. Barrel Part No.	Top Stop Set.	Total Travel	Cup Part No.	Cup Holes No. - Size x Angle	Approx. Injector Orifice	3375317 Delivery mm^3 /Stroke	Cam
3076702	3076126	3053422	0.393	-.493	3279720	10-.0095x 10deg	N/A	275	426
3076703	3076126	3053422	0.393	0.493	3279720	10-.0095x 10deg	0.025	200	426
3076704	3076126	3053422	0.393	0.493	3279747	10-.0100x 10deg	0.033	225	426
3077714	3076127	3063012	0.302	0.404	3042713	10-.0085x 10deg	0.03	285.5	310
3077715	3076125	3053420	0.302	0.404	3077716	10-.0090x 10deg	0.038	365	310
3077760	3076125	3053420	0.302	0.404	3042713	10-.0085x 10deg	0.03	285.5	310
3627922	3076126	3053422	0.393	0.493	3279747	10-.0100x 10deg	0.025	190	426

Note : To meet compliance with emissions documentation requirements, the engine dataplate **must** be stamped with FF177 when this engine upgrade is performed on engines with CPL of 0801. The dataplate only needs to be stamped for engines with an 0801 CPL.

Document History

Date	Details
2011-4-29	Module Created